

1.5 Classwork: Applications

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$ area of a circle = πr^2

speed of light $c = 3.00 \times 10^8$ m/s 1 km = 1000 m 1 m = 100 cm
1 h = 3600 s 1 min = 60 s 1 yr $\approx$ 365 days 1 metric ton = 1000 kg
1 m³ = 1000 L 1 L of water has a mass of 1 kg 1 gal $\approx$ 3.785 L

1. During a heavy storm, 2.5 cm of rain falls on Manhattan. Treat Manhattan as a rectangle 20 km long and 3.0 km wide.

(a) What volume of water fell, in cubic meters? Convert each length to meters first.

$$\begin{aligned} 1 \text{ km} &= 1000 \text{ m} & 20 \text{ km} &= 20,000 \text{ m} & V &= l \cdot w \cdot h \\ 1 \text{ cm} &= 0.01 \text{ m} & 3.0 \text{ km} &= 3,000 \text{ m} & &= 20,000 \text{ m} \times 3,000 \text{ m} \times 0.025 \text{ m} \\ & & 2.5 \text{ cm} &= 0.025 \text{ m} & &= 1,500,000 \text{ m}^3 \\ & & & & &= 1.5 \times 10^6 \text{ m}^3 \end{aligned}$$

(b) What is the mass of this water, in kilograms and in metric tons?

$$\begin{aligned} 1.5 \times 10^6 \text{ m}^3 &\times \frac{1000 \text{ L}}{1 \text{ m}^3} \times \frac{1 \text{ kg}}{1 \text{ L}} = 1.5 \times 10^9 \text{ kg} \text{ (also, Liters)} \\ 1 \text{ ton} &= 1000 \text{ kg} \quad 1.5 \times 10^9 \text{ kg} = 1.5 \times 10^6 \text{ metric tons} \end{aligned}$$

(c) How many gallons is it? New York City uses about 1 billion gallons of water a day — how does the storm compare?

$$\begin{aligned} 1.5 \times 10^9 \text{ L} &\times \frac{1 \text{ gallon}}{3.785 \text{ L}} = 3.963... \times 10^8 \text{ gallons} \\ &\approx 4.0 \times 10^8 \text{ gallons} \\ &= 400,000,000 \text{ gallons} \\ &\text{Less than half of a billion} \\ &\text{(40\% of a day's use)} \end{aligned}$$

1890 - Lord Rayleigh accurate experiment
following Franklin

2. In 1773 Benjamin Franklin poured a teaspoon of oil on a pond and watched it spread into a thin film. Suppose 1.0 L of oil spreads over a calm lake until the slick is just one molecule thick, about 2×10^{-10} m.

(a) What is the volume of the oil in cubic meters?

$$V = 1 \text{ L} \times \frac{1 \text{ m}^3}{1000 \text{ L}} = 0.001 \text{ m}^3 = 1 \times 10^{-3} \text{ m}^3$$

(b) The volume of the slick equals its area times its thickness. What is the area of the slick, in square meters?

$$V = A \cdot 2 \times 10^{-10} \text{ m} = 1 \times 10^{-3} \text{ m}^3$$

$$A = \frac{1 \times 10^{-3} \text{ m}^3}{2 \times 10^{-10} \text{ m}} = 5 \times 10^6 \text{ m}^2$$

(c) If the slick is a circle, estimate its diameter.

$$A_{\text{circle}} = \pi r^2 = 5 \times 10^6 \text{ m}^2$$

$$r = \sqrt{\frac{5 \times 10^6 \text{ m}^2}{\pi}} = 1261.5... \text{ m} = 2523... \text{ m} \approx 3,000 \text{ m} = 3 \text{ km}$$

$d = 2r = 2(1261.5...) \text{ m} = 2523... \text{ m} \approx 3,000 \text{ m} = 3 \text{ km}$

3. The distance around the Earth at the equator is 40,075 km. Suppose you could walk it at 5.0 km/h, walking 12 hours each day.

(a) How many days would the trip take?

$$40,075 \text{ km} \times \frac{1 \text{ hour}}{5.0 \text{ km}} \times \frac{1 \text{ day}}{12 \text{ h}} = 667.91... \text{ days} \approx 670 \text{ days}$$

(b) About how many years is that?

$$667.91... \text{ days} \times \frac{1 \text{ yr}}{365 \text{ days}} = 1.829... \text{ yr} \approx 1.8 \text{ years}$$

4. The Sun is 1.496×10^8 km from the Earth. How many minutes does it take sunlight to reach us? Write the conversion as a single chain of fractions.

$$1.496 \times 10^8 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ second}}{3.00 \times 10^8 \text{ m}} \times \frac{1 \text{ m}}{60 \text{ s}} =$$

$$8.311... \text{ min} \approx 8.31 \text{ minutes}$$